

Grammatical features of successful exam writing: a multi-dimensional comparison of human and LLM texts

This study looks at the grammatical correlates of writing success in university entrance composition exams and considers how large language models can be used as writing partners in exam preparation. Previous research has shown that linguistic features such as syntactic complexity and lexical sophistication are associated with writing quality (e.g. Crossley et al., 2010). However, such studies tend to rely on overall pre-defined measures rather than identifying the specific grammatical features that characterize exam writing. To address this issue and determine the extent to which different LLMs can provide human-like texts that meet entrance exam requirements, we conduct a corpus-based study on the grammatical features associated with composition writing. The human-authored corpus consists of 500 compositions written for the entrance exam of a major state university in Brazil, where failure in the composition component results in overall exam failure. The corpus was divided into passing (P) and failing (F) texts, with the latter being rewritten submitted by two LLMs (GPT 5.2 and Gemini 3.1) according to the official six-criterion rating scheme. All texts were tagged with the PALAVRAS Parser and counts the linguistic features were taken by a specialized script, which were then submitted to a Multi-Dimensional Analysis (Berber Sardinha & Veirano Pinto, 2019; Biber, 1988; Veirano Pinto et al., 2026). The resulting dimensions were interpreted and the LLM-generated texts compared to those of human-authored passing texts. Results indicate differences in grammatical resources between successful human writing and successful LLM texts, with implications for exam preparation.

Keywords: University entrance composition exams, LLM-generated texts, Human-authored texts, Grammatical features, Multi-dimensional analysis.

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